
Interdisciplinary Studies, Grade 11, Open

(IDC3O/IDP3O)

This course will help students combine the skills required for and knowledge of different subjects and disciplines to solve problems, make decisions, create personal meaning, and present findings beyond the scope of a single subject or discipline. Through individual and collaborative inquiry and research, students will analyse the connections among diverse subjects and disciplines; develop information literacy skills in analysing, selecting, evaluating, and communicating information; and become aware of a variety of resources and viewpoints on contemporary issues. They will also examine their own learning styles, relate their inquiries and research to real-life situations, and investigate career opportunities in new disciplines.

Prerequisites: for IDC3O, none; for IDP3O, the prerequisite for each of the courses in the package

Theory and Foundation

Overall Expectations

By the end of this course, students will:

- demonstrate an understanding of the key ideas and issues related to each of the subjects or disciplines studied;
- demonstrate an understanding of the different structures and organization of each of the subjects or disciplines studied;
- demonstrate an understanding of the different perspectives and approaches used in each of the subjects or disciplines studied;
- demonstrate the skills and strategies used to develop interdisciplinary products and activities.

Specific Expectations

Ideas and Issues

By the end of this course, students will:

- identify the fundamental ideas and issues that characterize each of the subjects or disciplines studied and critically analyse how the ideas and issues interrelate (e.g., for an interdisciplinary studies course on business ethics: the ethical use of information technology, the impact of international economic associations, ethical issues in global marketing, and the impact of entrepreneurs on a community; for an interdisciplinary course on ecotourism: geography concepts related to ways in which regional factors influence human movement and interaction, and business studies concepts related to ways in which changes in demographic and geographic characteristics influence potential tourist markets);
- describe and critically analyse key issues and themes that have interdisciplinary elements and connections in the subjects or disciplines studied (e.g., the application of mathematical concepts of patterning and sequence in music and in dance; the impact of leadership styles and personal management skills on successful tourism and hospitality ventures; the influence of political, economic, and social factors in organized sport);
- identify and describe the features and roles of information literacy in each of the subjects or disciplines studied (e.g., by investigating the effect of the “digital economy” on human relationships and the world of work, by examining how global data sharing can assist the sustainable development of natural resources);
- identify and describe, with particular reference to each of the subjects or disciplines studied, the principles and practices regarding the safe, ethical, and legal use of information and information technologies (e.g., “netiquette”, personal privacy and security, copyright, software user agreements).

Structures and Organization

By the end of this course, students will:

- analyse and describe ways in which each of the subjects or disciplines studied are structured or organized (e.g., “How do the natural sciences and human sciences differ in the ways in which they classify human behaviour?”, “How distinct are the different forms and genres of literary expression?”, “How successfully does the Dewey Decimal System classify knowledge?”);
- explain the importance of organizing and storing information and resources to each of the subjects or disciplines studied (e.g., to record the development of knowledge,

to provide access to municipal bylaws and provincial legislation, to advance scientific research around the world);

- identify and describe the ways in which information is classified, organized, stored, and retrieved in selected print and electronic forms used in each of the subjects or disciplines studied (e.g., in fictional and instructional books and periodicals, audio and multimedia forms, census and statistical reports, personal and community web pages);
- identify and describe the features of a variety of information systems (e.g., local and national online public access catalogues, search engines and directories, web portals – websites or services that offer a broad array of resources and services, such as e-mail, forums, search engines, and online retail services).

Perspectives and Approaches

By the end of this course, students will:

- analyse and describe how each of the subjects or disciplines studied views the role of personal experience in gaining knowledge (e.g., “How does our personal experience affect our perception of things?”, “What personal responsibility does claiming to ‘know’ something carry for the ‘knower’?”, “What roles do culture and language play in our perceptions, understandings, and beliefs?”);
- analyse and describe the different perspectives of each of the disciplines involved in an enterprise (e.g., the engineer’s interest in making roads safe and the sociologist’s interest in who benefits from the new roads; the journalist’s interest in reporting a news story and a citizen’s interest in maintaining privacy);
- identify and describe the different approaches used to investigate topics in interdisciplinary work (e.g., statistical

analyses in sociology, ethnographic studies in anthropology, case studies in business studies);

- identify and describe the features and applications of the systems-thinking approach (e.g., the identification of external and internal factors, the application of structured problem-solving strategies and models).

Skills and Strategies

By the end of this course, students will:

- demonstrate an understanding of the collaborative attitudes and skills that are valuable when researching and creating interdisciplinary products and activities (e.g., to solicit feedback on a business plan and then revise it by elaborating on its ideas; to plan a hospitality event using divergent thinking within the group to brainstorm new ideas and convergent thinking to evaluate the group’s ideas systematically; to use enterprise, flexibility, and empathy to help resolve civic disputes);
- identify the conventions of language used in each of the different subjects or disciplines studied (e.g., common and less-common terminology, recurrent idioms and expressions);
- critically analyse and demonstrate the ability to apply a variety of critical- and creative-thinking strategies and models (e.g., brainstorming techniques, decision-making processes, Edward de Bono’s lateral-thinking models) to help develop effective interdisciplinary products or activities;
- demonstrate an understanding of how a variety of information technologies are used to support interdisciplinary work (e.g., electronic spreadsheets to manipulate data and present information, web-based simulations to test scientific hypotheses, electronic graphic organizers to develop and link ideas).

Processes and Methods of Research

Overall Expectations

By the end of this course, students will:

- be able to plan for research, using a variety of strategies and technologies;
- be able to access appropriate resources, using a variety of research strategies and technologies;
- be able to process information, using a variety of research strategies and technologies;
- be able to assess and extend their research skills to present their findings and solve problems.

Specific Expectations

Preparing for Research

By the end of this course, students will:

- demonstrate an understanding of the purposes and types of research used in each of the subjects or disciplines studied (e.g., by distinguishing between research done in the arts and that done in the sciences, by comparing methods of research in traditional and in alternative medicine) and describe examples of effective local research projects;
- demonstrate an understanding of the skills and attitudes required for research in each of the subjects or disciplines studied (e.g., by using information-management skills such as recording, note taking, and outlining; by valuing their own learning styles and research methods and those of others; by using methods and stages of inquiry and research);
- formulate questions for a variety of purposes in interdisciplinary research (e.g., to explore a topic and theme, to identify gaps in prior knowledge, to broaden individual perspectives);
- develop and manage personal plans for interdisciplinary research, products, and activities, using a variety of print and electronic resources and organizers (e.g., using tables and charts to record their research needs; using sequence diagrams and time-management applications to check action

taken and decisions made; using graphic organizers or webbing applications to analyse feedback from peers).

Accessing Resources

By the end of this course, students will:

- identify and describe physical and virtual sources of information (print, electronic, and mass media) that are relevant to interdisciplinary research (e.g., school and public libraries, national libraries and government archives, resources on the Internet, experts from museums and galleries);
- identify the conventions used in databases, catalogues, and indexes to organize information (e.g., subject headings and descriptors, cross-references and see-also references, call numbers), and apply this knowledge to locate relevant resources for interdisciplinary research, using a variety of search strategies and features (e.g., Boolean operators, key-word searches, standard and advanced features of search engines);
- locate relevant resources for interdisciplinary research, using a variety of print or electronic (online) reference material, indexes, and databases (e.g., specialized encyclopaedias and dictionaries, online periodical indexes and full-text databases, almanacs and yearbooks);

- select relevant information for interdisciplinary research within resources, using a variety of reading and critical-thinking strategies (e.g., scanning tables of contents, indexes, and chapter summaries for their organization and treatment of ideas; taking notes on key ideas; recording experimental data).

Processing Information

By the end of this course, students will:

- analyse and evaluate information from a variety of print, electronic, and mass media resources according to specific criteria, including the clarity, interest, and accuracy of the information (e.g., “How clearly is the purpose of the theological anthology stated in the preface?”, “Would analysis of art works on the gallery website engage a popular audience?”, “Are the historical facts regarding the Holocaust stated accurately in the film documentary?”);
- identify and critically analyse ideas, arguments, bias, and stereotyping found in resources, using a variety of strategies (e.g., defining terms central to an argument, identifying relationships among data in a work of interdisciplinary studies, conducting an Internet search to investigate divergent arguments from a range of sources);
- record, sort, and organize information found in resources related to each of the subjects or disciplines studied, using a variety of print and electronic organizers (e.g., charts, relational databases, graphic organizers);
- identify the reasons for acknowledging sources and use an accepted form of documentation to record sources of information;

- synthesize findings from interdisciplinary research, using a variety of strategies and technologies (e.g., describing relationships among the data and information found in a variety of resources, showing relationships among data using graphic organizers, delivering a slide presentation on the social impact of amateur sport in Canada based on information from print and online encyclopaedias related to different disciplines).

Assessing and Extending Research

By the end of this course, students will:

- assess the quality and effectiveness of their research in comparison to similar research conducted by others (e.g., by comparing conclusions reached about common examples, by recognizing the difference between professional and student work);
- assess the effectiveness of their research in meeting the original information requirements and their research plans (e.g., “How could I adjust my method of gathering investment statistics to improve my financial results?”, “What new course of action is required in the local community’s ecological project?”);
- develop and apply effective criteria for assessing the quality of their interdisciplinary research (e.g., by identifying how well they organized data and synthesized information to come up with new ideas);
- identify and describe possible topics and real-life applications for subsequent interdisciplinary research activities.

Implementation, Evaluation, Impacts, and Consequences

Overall Expectations

By the end of this course, students will:

- implement and communicate information about interdisciplinary endeavours, using a variety of methods and strategies;
- evaluate the quality of interdisciplinary endeavours, using a variety of strategies;
- analyse and describe the impact on society of interdisciplinary approaches and solutions to real-life situations;
- analyse and describe ways in which interdisciplinary skills relate to personal development and careers.

Specific Expectations

Implementation and Communication

By the end of this course, students will:

- create interdisciplinary products based on their own plans or designs, independently or as members of a team (e.g., a series of fashion designs for ten years in the future using traditional or digital illustrations, a health and safety plan in response to a customer survey);
- demonstrate the ability to communicate and present information effectively, using a variety of methods and forms (e.g., written and oral reports, dramatic presentations, annotated graphs and charts);
- demonstrate an understanding of the visual and textual ways of representing information in each of the subjects or disciplines studied (e.g., nutritional charts; graphs and diagrams; different geographic, political, historical, and cultural maps; stock market indexes and tables);
- identify and describe ways in which information in each of the subjects or disciplines studied is shaped and communicated by the media used (e.g., print as contrasted with multimedia; text as contrasted with visual representations; still versus moving images).

Evaluation

By the end of this course, students will:

- monitor the effectiveness of the plans for their interdisciplinary products or activities using appropriate strategies (e.g., developing criteria to measure effective implementation, checking progress according to a time line, providing progress reports on action taken and decisions made during the process);
- develop and apply effective criteria for evaluating the quality of their interdisciplinary products and activities (e.g., to identify how well their products and activities demonstrate their ability to manage data and present ideas, to synthesize ideas and approaches, or to create innovative and original explorations or solutions);
- evaluate and record the effectiveness of their interdisciplinary activities and projects, including their research sources, methods, findings, and plans (e.g., by identifying how well their activities and projects reflected characteristic methods and approaches of each of the subjects and disciplines under study; by revising their plans as problems and solutions arose; by analysing how perspectives were shaped by the sources of information used);

- evaluate the effectiveness of the collaborative strategies they used in planning and implementing interdisciplinary products and activities (e.g., by identifying how well they valued diversity, respected individual positions on an issue, considered alternatives and new connections, and actively sought feedback and constructive criticism).

Impacts and Innovations

By the end of this course, students will:

- describe and critically analyse contemporary examples of interdisciplinary products and activities that provide innovative approaches and solutions to a variety of real-life situations in the local community (e.g., the coordination of local transportation systems, the delivery of services through e-commerce and e-government, the development of community health and recreation facilities);
- identify and describe new and emerging interdisciplinary branches of learning and fields of study (e.g., biotechnology, computer animation, forensic pathology, environmental law, knowledge management);
- research, analyse, and describe the personal and social impacts of significant information-related achievements (e.g., the development of world alphabets, the evolution of paper and books, the evolution of the moving image, the creation of digital text);
- plan, conduct, and present independent interdisciplinary research, with particular reference to each of the subjects or disciplines studied, on the potential social, political, and economic impacts of emerging information technologies (e.g., the inequitable access to and ability to use technology – the “digital divide”; the impact of information systems on personal freedom; the development of technology for the disabled).

Personal and Career Development

By the end of this course, students will:

- demonstrate an understanding of selected interdisciplinary texts related to each of the subjects or disciplines studied and identify significant titles for future study;
- identify their personal information skills and those skills that require development if they are to achieve success in interdisciplinary studies (e.g., their ability to identify information needs, to locate appropriate resources, to resolve conflict within a research team, and to plan new research);
- research the importance of effective collaborative and communication skills in interdisciplinary careers related to the subjects and disciplines under study (e.g., in oceanography: building consensus among national governments or coordinating research methods);
- identify and describe employment opportunities within the local community in interdisciplinary fields related to the subjects or disciplines under study (e.g., by conducting interviews with practitioners to record changes in their fields, analysing online job descriptions for interdisciplinary connections, comparing archival and contemporary classified advertisements to determine employment opportunities that are increasing and those that are decreasing).